

A custom induction furnace for high-temperature materials processing

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An abstract of fewer than 250 words summarizing the instrument, its novel aspects, and representative performance results goes here.

I. INTRODUCTION

Introduce the motivation for the instrument and survey prior work.¹ Citations use `\cite{key}` with entries in `references.bib`.

II. INSTRUMENT DESIGN

Describe the furnace design: induction power supply, coil and crucible geometry, vacuum system, gas handling, and instrumentation.

III. PERFORMANCE

Present representative measurements demonstrating the capabilities of the instrument.

IV. CONCLUSION

Summarize the instrument capabilities and outlook.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

¹Placeholder reference. Replace with real citations, e.g. the RSI example article: J. Author *et al.*, “Automated rapid cooling of high-temperature vacuum furnaces,” *Rev. Sci. Instrum.* **97**, 063303 (2026).

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